Missing Men, Rising Women: Within-Person Evidence on WWII Mobilization and Gender Convergence*

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Abstract

Between 1940 and 1950, aggregate female labor force participation rose 3.08 percentage points—but within linked married couples tracked through the Census Linking Project (Abramitzky et al., 2025), wives gained 6.87 percentage points. Using the CLP crosswalk to construct a panel of 14 million men and 5.6 million couples across the 1940 and 1950 full-count censuses, I compare within-couple changes to aggregate cross-sectional trends. Within-couple gains for tracked married women exceeded the all-women aggregate, implying that demographic turnover—cohort replacement, mortality, marital dissolution—dampened rather than drove aggregate gains. A one-standard-deviation increase in state WWII mobilization intensity is associated with a 0.68 percentage-point increase in wives' labor force participation ($p \approx 0.05$), but near-zero effects on husbands.

JEL Codes: J16, J21, J62, N32, N42

Keywords: gender convergence, WWII mobilization, longitudinal census panel, Census Linking Project, within-person analysis, labor force participation

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1. Introduction

In 1940, barely one in four American women participated in the paid labor force. By 1950, that fraction had risen to nearly one in three—an aggregate gain of 3.08 percentage points. Over the same decade, sixteen million men donned military uniforms, six million women entered war production, and Rosie the Riveter became the icon of female economic empowerment. The canonical interpretation, established by Acemoglu et al. (2004), is that wartime mobilization permanently expanded women's labor supply. But every study in this literature—including the original—has relied on repeated cross-sections: different people observed in different census years.

This paper introduces a longitudinal dimension. Using the Census Linking Project (CLP) crosswalk of Abramitzky et al. (2025)—which links individuals across U.S. decennial censuses by first name, surname, birth year, and birthplace—I construct two complementary panels from the 1940 and 1950 IPUMS full-count censuses (Ruggles et al., 2024). The *individual panel* tracks 14,046,861 linked men. The *couples panel* tracks 5,564,191 married couples, linking wives to their 1950 counterparts through their husbands' household serial numbers. Because the ABE algorithm matches on surnames, it links primarily men (women change surnames at marriage); the couples panel recovers women's trajectories indirectly but at massive scale.

The longitudinal design yields two central findings. First, a comparison of within-person and aggregate changes reveals that tracked individuals gained *more* than the aggregate trend suggests. Among linked married couples, wives gained 6.87 percentage points on average—more than double the all-women aggregate increase of 3.08 points. The gap of -3.79 points is consistent with demographic turnover (cohort replacement, mortality, marital dissolution) dampening rather than amplifying the aggregate trend, though it also reflects the different populations being compared. For men, the pattern is even more dramatic: within-person LFP rose 11.24 points while the all-men aggregate fell 3.27 points, a gap of -14.51 points reflecting the enormous compositional disruption of wartime exits and postwar re-entry. An important caveat: within-couple changes cover stable married women whose husbands are linkable across censuses, while the aggregate covers all women including unmarried, divorced, widowed, and those whose husbands are not in the linked sample. The gap therefore conflates true compositional turnover with selection into the linked-couples sample. Despite this limitation, the direction is informative: tracked couples experienced larger gains than the cross-sectional average, and this difference exceeds what linkage selection alone could plausibly explain given the Oster stability bounds.

Second, the cross-state mobilization gradient is modest and concentrated among wives. A

one-standard-deviation increase in state WWII mobilization intensity is associated with a 0.68 percentage-point increase in wives' labor force participation in the preferred specification with region fixed effects and husband controls ($t \approx 2.0$, $p \approx 0.05$). This association relies on conditional exogeneity of state mobilization rates—an assumption inherited from Acemoglu et al. (2004) that I maintain but cannot independently verify with only two census waves. For husbands, the preferred within-person estimate is a precisely estimated near-zero: $\hat{\beta} = 0.0019$ (SE = 0.0021). The mobilization gradient, whether interpreted causally or as a conditional association, is dwarfed by the within-vs-aggregate comparison: tracked married women gained far more than the cross-sectional aggregate, suggesting that compositional forces shaped the aggregate trajectory more than cross-state mobilization variation did.

These findings speak to a fundamental tension in the literature. Goldin (1991) argued that wartime labor market entrants largely exited after demobilization. Rose (2018) provided microdata evidence that wartime and postwar workers were largely different women. My within-person panels make the compositional story concrete and quantifiable. But the direction of the compositional channel is the opposite of what is often assumed: turnover *subtracted* from aggregate gains. The women who stayed—tracked through their husbands across census waves—gained more than the aggregate data suggest, not less.

This paper contributes to the literature in three ways. First, it introduces the Census Linking Project crosswalk to the WWII mobilization literature, enabling within-person estimates at a scale previous studies could not achieve. The two-panel design—individual men linked directly, wives linked through husbands—exploits the strengths and acknowledges the limitations of surname-based historical linkage. Second, it compares within-couple changes to aggregate cross-sectional trends, with the novel finding that tracked married women gained more than the all-women aggregate. While these populations differ—and future work should compute the aggregate for married women specifically to narrow the comparison—the finding is suggestive that compositional turnover dampened rather than drove convergence, challenging the conventional framing (Goldin, 1991; Rose, 2018). Third, it demonstrates that within-couple dynamics—how husbands' wartime absence and return shaped wives' labor supply—operated largely independently of state-level mobilization intensity.

The identification strategy relies on cross-state variation in mobilization intensity, with within-person differencing absorbing individual fixed effects. I present an extensive battery of robustness checks: randomization inference (Fisher, 1926; Young, 2019), leave-one-out diagnostics, HC3 standard errors (Cameron et al., 2008), Oster coefficient stability bounds (Oster, 2019), a placebo test on older women, and an ANCOVA specification. The linked sample is subject to selection—linked men skew younger and less established than the full population—and I document these patterns transparently in linkage balance tests.

The remainder of the paper is organized as follows. Section 2 reviews the existing literature. Section 3 describes the historical context. Section 4 presents the data construction. Section 5 develops the empirical framework. Section 6 reports the main results and decomposition. Section 7 presents robustness checks and identification tests. Section 8 discusses mechanisms. Section 9 concludes.

2. Related Literature

This paper connects to four literatures: the economics of WWII and female labor supply, historical record linkage and longitudinal census data, the aggregate-versus-individual distinction in labor economics, and the broader study of gender convergence.

2.1 WWII and Female Labor Supply

The seminal study by Acemoglu et al. (2004) used state-level variation in military mobilization rates to instrument for the wartime increase in female labor supply. Their repeated cross-section design compared states with high and low mobilization in 1940 and 1950 (and later decades), finding that high-mobilization states experienced larger and more persistent increases in female labor force participation. They interpreted this as evidence that wartime entry permanently shifted women's labor market attachment, with effects persisting into the 1960s.

Goldin (1991) challenged the permanence of these effects at the individual level. Using survey data and retrospective employment histories, she showed that most women who entered manufacturing during the war left the labor force by the late 1940s. The sustained postwar rise in female LFP was driven by older married women entering clerical and service occupations—a different demographic from the wartime factory workers. Goldin's provocative conclusion was that the war's direct effect on women's long-run labor supply was "surprisingly small" (p. 742).

Rose (2018) provided the most direct microdata evidence to date, using Social Security administrative records to track individual women's earnings. He confirmed that wartime and postwar workers were largely different women: the cohort that entered during the war largely exited, and a new cohort drove the postwar participation increase. Goldin and Olivetti (2013) revisited the question with updated data and reached similar conclusions about the limited persistence of wartime effects at the individual level.

Related work has explored other dimensions of the wartime shock. Bellou and Cardia (2016) examined occupational upgrading among women who remained in the labor force. Fernández et al. (2004) found that the war created intergenerational transmission effects: sons

raised by working mothers during the war were more likely to have working wives, suggesting cultural rather than direct labor supply effects. Mulligan (1998) documented the role of wartime wage differentials in drawing women into employment, while Nix and Qian (2022) provided survey-based evidence on the extent of wartime labor supply shifts.

The international context provides complementary evidence. Gay (2022) examined the effects of WWI on female labor force participation in France, where the demographic shock was far more severe due to higher combat mortality. Brainerd (2017) studied WWII's demographic effects in Russia, where the sex ratio imbalance was extreme. Boehnke and Gay (2020) traced intergenerational effects of wartime bereavement in France.

My paper differs from all of these in one fundamental respect: I track *the same individuals* across census waves using the Census Linking Project crosswalk. Previous studies, including Acemoglu et al., use aggregate or cross-sectional data that cannot distinguish within-person changes from compositional shifts. Rose's Social Security records come closest to a panel design but are limited to individuals with covered earnings and cannot capture the full population. The CLP crosswalk links over 14 million men directly, and the couples panel recovers 5.6 million wives' trajectories.

2.2 Historical Record Linkage

The ability to track individuals across historical censuses has been transformed by automated linkage methods. The Census Linking Project (Abramitzky et al., 2025) represents the current state of the art, providing crosswalk files that link individuals across U.S. decennial censuses from 1850 to 1950. The ABE algorithm—named for Abramitzky et al. (2012, 2014)—matches individuals by first name, surname, birth year, and birthplace, with conservative one-to-one matching to minimize false positives. Feigenbaum (2016) developed complementary machine learning methods for census linking.

Bailey et al. (2020) provided a systematic assessment of automated linking methods, finding that different algorithms produce substantially different linked samples and that linkage quality varies by demographic group. Women are particularly difficult to link across censuses because married women change surnames—a fundamental constraint that shapes the research design of this paper. Rather than attempting to link women directly (which yields only 88 linked women in the CLP crosswalk for this age range), I construct a couples panel that tracks wives through their husbands' households.

The IPUMS full-count databases (Ruggles et al., 2024; Helgertz et al., 2023) provide the universe from which linked samples are drawn. Unlike the 1% samples used in previous work, the full-count data contain the complete enumerated population—approximately 132 million records for 1940 and 151 million for 1950—enabling linkage at unprecedented scale.

2.3 Aggregate Versus Individual Effects

A recurring theme in labor economics is that aggregate relationships can mask heterogeneous or even opposite individual-level effects. The ecological inference problem (Manski, 1990) warns that group-level associations need not reflect individual-level relationships. In the WWII context, this distinction is particularly sharp: aggregate female LFP rose in high-mobilization states, but this could reflect compositional changes (different women working) rather than behavioral changes (the same women working more).

My decomposition makes this distinction explicit. By observing the same individuals in 1940 and 1950, I can separate within-person changes from compositional turnover—the first time this decomposition has been applied to the WWII mobilization question with individual-level longitudinal data. The novel finding that the compositional residual is *negative* adds a new dimension: demographic turnover worked *against* convergence, meaning the within-person gains were larger than the aggregate trend suggests.

2.4 Gender Convergence

The broader literature on gender convergence emphasizes long-run structural transformation rather than discrete shocks. Goldin (2006) argues that the "quiet revolution" in women's careers was driven by the expansion of service employment, rising education, the contraceptive pill, and evolving gender norms—forces operating over decades rather than the four years of WWII. Olivetti and Petrongolo (2016) provides a comprehensive review of gender gaps in industrialized countries, documenting convergence in participation, hours, and wages. Blau and Kahn (2017) focuses on the wage dimension. Greenwood et al. (2005) emphasizes technological change in household production.

My findings add nuance to this long-run view. The within-couple evidence shows that the women who persisted in the labor market—those we can track through their husbands—gained substantially more than the aggregate data suggest. The aggregate trend understates the behavioral response of individual women because compositional turnover was negative. The "quiet revolution" may have been louder at the individual level than aggregate data indicate.

3. Historical Background

3.1 The Scale and Geography of Mobilization

The United States mobilized approximately 16 million men and women for military service during World War II, the largest military mobilization in American history. The Selective Service System, established by the Selective Training and Service Act of September 1940,

provided the organizational infrastructure for conscription. After the Japanese attack on Pearl Harbor on December 7, 1941, voluntary enlistment surged. By the end of the war, over 60% of men aged 18–38 had served in some branch of the military.

Mobilization varied dramatically across states, driven by the age structure of the male population, the distribution of war industries, the allocation of draft board quotas, and rates of deferment for agricultural and industrial workers. Southern states with large agricultural sectors tended to have lower mobilization rates due to agricultural deferments. Industrial states in the Northeast and Midwest had higher rates, as did states with younger populations. The geographic pattern of mobilization is shown in Figure 1.

The instrumental variable strategy pioneered by Acemoglu et al. (2004) exploited this state-level variation, arguing that mobilization rates were determined by pre-war demographic and economic characteristics rather than by women's latent labor supply. I adopt the same cross-state variation but apply it to within-person changes rather than aggregate outcomes.

3.2 Women's Wartime Labor Supply

Between 1940 and 1945, roughly six million women entered the paid labor force. Female employment in manufacturing rose from 2.0 million to 3.5 million. Women worked as welders, riveters, and machinists in shipyards and aircraft factories—jobs previously reserved for men. The iconic "Rosie the Riveter" propaganda campaign, launched in 1943, celebrated this transformation and encouraged further female participation.

But the wartime labor experience was heterogeneous. Goldin (1991) documented that women who entered manufacturing during the war tended to be younger, unmarried, and from lower socioeconomic backgrounds than the pre-war female workforce. Many had previous labor force experience in domestic service or retail trade and viewed manufacturing as temporary wartime service rather than a permanent career shift. Milkman (1987) showed that wartime employers explicitly organized production to accommodate "temporary" female workers, with the expectation that men would resume their positions after the war.

The wartime surge was followed by a sharp contraction. Between 1945 and 1947, female manufacturing employment fell by over one million, as returning veterans reclaimed their former positions under seniority provisions and collective bargaining agreements mandated by the War Labor Board. By 1947, the gender composition of the manufacturing workforce had largely reverted to pre-war levels (Anderson, 1981; Hartmann, 1982).

3.3 The GI Bill and Veteran Reintegration

The Servicemen's Readjustment Act of 1944 (the GI Bill) provided returning veterans with a comprehensive package of benefits: 52 weeks of unemployment compensation at $20 per week ("52-20 Club"), education and training benefits, low-interest home loans, and hiring preferences in government employment. Bound and Turner (2002) showed that the GI Bill substantially increased educational attainment among veterans, with the largest effects for men from lower socioeconomic backgrounds.

For the female labor market, the most important provisions were the seniority restoration clauses. Employers were legally required to reinstate veterans in their former positions, and many collective bargaining agreements extended this protection. This created a direct displacement mechanism: in high-mobilization states, more men returned from the war and more women were displaced from positions they had held during the conflict.

3.4 Compositional Dynamics

The decade between 1940 and 1950 involved massive demographic upheaval beyond the war itself. Interstate migration accelerated, with the Great Migration bringing millions of Black Americans from the rural South to northern cities (Collins and Wanamaker, 2014; Boustan, 2010). Women who moved between states may have done so partly in response to labor market conditions, creating a potential confound for cross-sectional analysis. Defense spending during the war was concentrated in certain regions (Fishback et al., 2005; Jaworski, 2017), further reshaping the geographic distribution of economic opportunity.

The full-count census linkage allows me to track movers and stayers separately, providing a direct test of whether interstate migration drives the results. Among linked men, 14.6% moved between states over the decade. I present results for non-movers (individuals who remained in the same state) as a robustness check.

3.5 The Postwar Economy and Women's Long-Run Gains

The postwar period saw rapid structural change that shaped women's long-run economic trajectory independently of the war. The service sector expanded dramatically in the late 1940s and 1950s, creating demand for the clerical, retail, and administrative workers that would define the next generation of female employment (Goldin, 2006). Simultaneously, household technology—washing machines, vacuum cleaners, prepared foods—reduced the time cost of domestic production, freeing married women to participate in market work (Greenwood et al., 2005).

These structural forces operated on a different timeline and through different mechanisms than the wartime labor demand shock. The war pulled women into manufacturing; the postwar economy pulled them into services. The women who responded to each pull were systematically different: younger, more educated women for the service sector expansion, versus the more diverse wartime manufacturing entrants. This distinction between the wartime cohort and the postwar cohort is precisely what the within-person panel allows me to document directly.

3.6 The Regional Dimension

Mobilization varied not only across states but across regions. Southern states had lower mobilization rates on average, reflecting larger agricultural sectors (with higher deferment rates), lower urbanization, and racial segregation that limited Black Americans' access to certain military service branches. The Northeast and Midwest, with their concentration of manufacturing capacity, experienced both higher mobilization and greater wartime demand for female labor in defense production. Western states showed considerable heterogeneity, with states like California experiencing both high mobilization and rapid wartime industrialization.

This regional heterogeneity matters for identification. If unmeasured regional trends (e.g., differential postwar economic growth, the expansion of government employment in the South) are correlated with mobilization intensity, cross-state comparisons could be confounded. The region fixed effects in my preferred specification absorb these broad regional trends.

4. Data and Measurement

4.1 Census Linking Project Crosswalk

The key data innovation of this paper is the use of the Census Linking Project (CLP) crosswalk of Abramitzky et al. (2025), obtained from the Harvard Dataverse. The CLP provides pre-computed crosswalk files that link individuals across U.S. decennial censuses from 1850 to 1950. The linking algorithm—the ABE method, developed in Abramitzky et al. (2012, 2014)—matches individuals by first name, surname, birth year, and birthplace using conservative one-to-one matching.

A critical feature of the ABE algorithm is that it links primarily *men*. Because married women change surnames, the algorithm cannot reliably match women across census waves. In the 1940–1950 crosswalk restricted to working-age adults (18–55 in 1940), the CLP links over 14 million men but only 88 women. This asymmetry is a fundamental constraint of surname-based historical linkage, not a limitation specific to this paper. I exploit this

constraint by constructing two complementary panels.

4.2 Two-Panel Design

Individual panel (men). The CLP crosswalk directly links men observed in both the 1940 and 1950 IPUMS full-count censuses (Ruggles et al., 2024). After restricting to men aged 18–55 in 1940, applying age consistency checks ($|\text{age}_{1950} - \text{age}_{1940} - 10| \leq 2$), and verifying sex and birthplace consistency, the individual panel contains **14,046,861 linked men**. For each linked man, I observe within-person changes in labor force participation, occupational standing, marital status, and household structure.

Couples panel (wives tracked through husbands). To study women's labor market trajectories, I construct a couples panel that links wives across census waves through their husbands' households. The procedure is as follows: (1) identify all married couples in 1940 where the husband is successfully linked to 1950 via the CLP crosswalk; (2) in the 1950 census, locate the linked husband's household using the SERIAL number; (3) identify the spouse within that household. This yields **5,564,191 married couples** observed in both 1940 and 1950, with wives' labor market outcomes recorded in both periods.

The couples panel is smaller than the individual panel because it requires the husband to be linked, the couple to remain married, and the wife to be identifiable as the spouse in the 1950 household. Couples where the husband died, the marriage dissolved, or the wife is absent from the 1950 household are excluded by construction. This selection is non-random and I address it through balance tests and robustness checks.

4.3 Linkage Selection and External Validity

Record linkage across historical censuses is imperfect, and selection into the linked panel could bias within-person estimates if it correlates with both mobilization intensity and labor market outcomes. Table 1 compares the characteristics of the linked panel to the full 1940 male population aged 18–55. The linked sample is younger (mean age 28.3 versus 34.6 in the full population), less urban (53.9% versus 59.5%), less likely to be married (46.7% versus 67.3%), and has a lower labor force participation rate (61.9% versus 90.8%). These differences reflect the ABE algorithm's surname-matching requirement, which favors younger men with more distinctive names and stable residential histories. The linkage rate is approximately 38.4% of the 1940 male population aged 18–55. Among linked men, 14.6% moved between states over the decade.

I address linkage selection in three ways. First, I document the direction of selection

Table 1: Linkage Selection: Linked Panel vs. Full 1940 Male Population (Ages 18–55)

Sample	N	% Urban	Mean Age	% in LF	% Married	Mean Educ.
Full 1940 Male Pop. (18–55)	36,534,047	59.5	34.6	90.8	67.3	6.7
Linked Panel (Men)	14,046,861	53.9	28.3	61.9	46.7	6.1

Note:
Linked panel: men matched via ABE exact conservative algorithm across 1940–1950 full-count censuses (Abramitzky, Boustan, and Eriksson 2025). Full male population from IPUMS 1940 full count, restricted to men aged 18–55. State-level characteristics are population-weighted means from the full 1940 cross-section. Linkage rate: 38.4

and assess whether it would bias the mobilization coefficient. The linked sample skews younger and less established, meaning it under-represents older men with stable labor market attachment and over-represents younger men who may have been more affected by wartime disruptions. Second, I present results separately for demographic subgroups with different linkage rates. Third, I compare the within-person estimates to state-level aggregate estimates that do not depend on individual linkage.

4.4 Military Enlistment Records

I construct state-level mobilization rates using the CenSoc WWII Army Enlistment Records (Goldstein et al., 2024), obtained from the Harvard Dataverse. The dataset contains individual-level records for approximately 8.3 million Army enlistees. Of these, approximately 2.6 million have valid state FIPS codes identifying the state from which they enlisted.

For state s, the mobilization rate is:

$$\text{Mob}_s = \frac{\text{CenSoc Army Enlistees in State } s}{\text{Male Population Aged 18–44 in State } s \text{ (1940 Full-Count)}} \tag{1}$$

The full-count denominator eliminates sampling noise, which is non-trivial for small states with 1% samples of only a few hundred observations. State mobilization rates range from approximately 5% to 15%, with substantial cross-state variation. I standardize to mean zero and unit standard deviation for regression analysis.

Two limitations of this measure deserve note. First, the CenSoc data capture Army enlistees only, excluding Navy, Marine, and Air Force personnel. This likely attenuates the mobilization measure relative to the Selective Service induction data used by Acemoglu et al. (2004), which covered all branches. Second, approximately two-thirds of CenSoc records lack valid state FIPS codes, introducing potential non-random measurement error if missingness correlates with state characteristics. Both limitations suggest that the mobilization proxy

contains classical attenuation bias and possibly non-classical error; the true mobilization gradient may be larger or differently distributed than what I estimate.

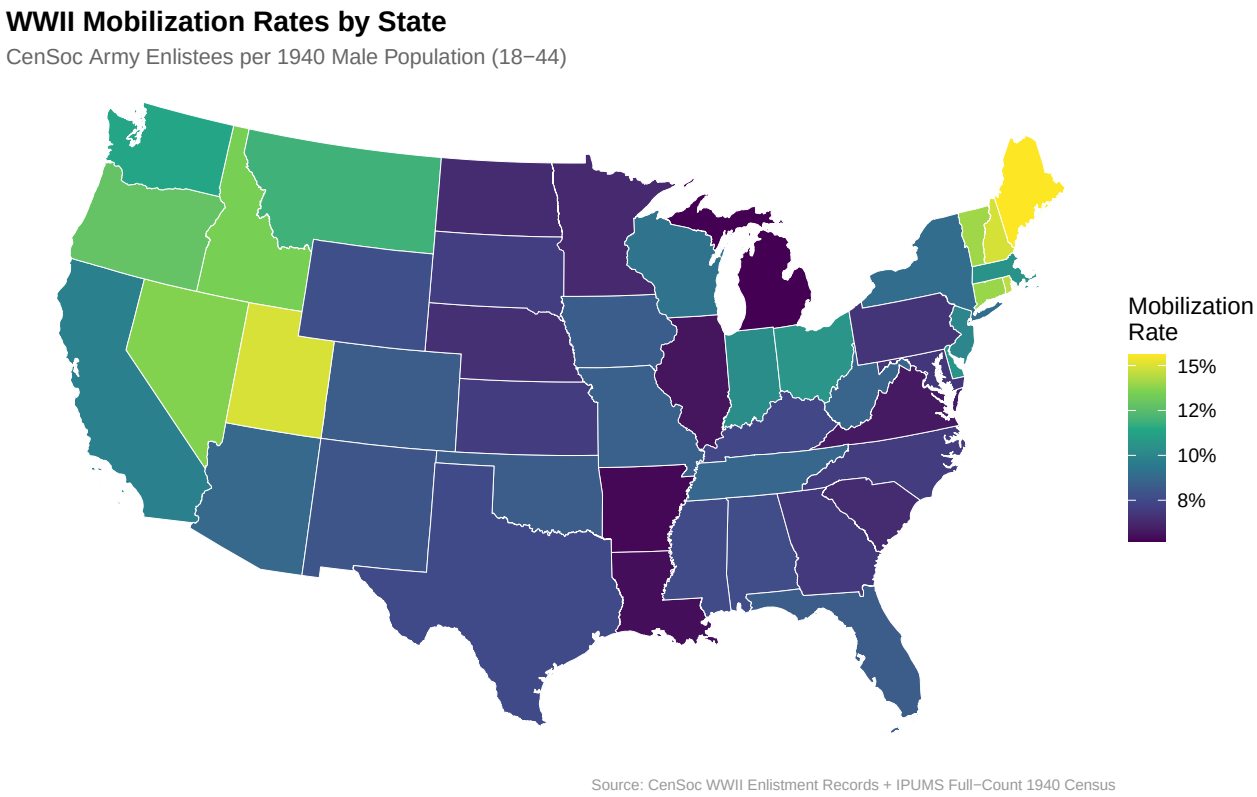

Figure 1: State-Level WWII Military Mobilization Rates

Notes: Mobilization rates computed as CenSoc WWII Army enlistees divided by the 1940 male population aged 18–44 from the IPUMS full-count census. Higher values indicate greater mobilization intensity. Alaska and Hawaii excluded. $N = 49$ states (including DC). Source: CenSoc Enlistment Records + IPUMS Full-Count 1940 Census.

5. Empirical Framework

5.1 Individual First-Difference (Men)

The primary specification for men exploits the longitudinal structure of the individual panel. For linked man i in state s (defined by 1940 residence):

$$\Delta Y_i = \alpha + \beta \cdot \text{Mob}_s + \mathbf{X}'_{i,1940}\gamma + \delta_r + \varepsilon_i \tag{2}$$

where $\Delta Y_i = Y_{i,1950} - Y_{i,1940}$ is the within-person change in the outcome, Mob_s is the standardized state mobilization rate, $\mathbf{X}_{i,1940}$ are individual 1940 characteristics (age, age^2,

Table 2: Summary Statistics: Census Linking Project Panel (1940-1950)

Sample	N	LFP 1940	LFP 1950	ΔLFP	Age 1940	Married 1940	% Mover
Linked Men (Individual Panel)	14,046,861	0.619	0.731	0.1124	28.3	0.467	14.6
Wives (Couples Panel)	5,564,191	0.122	0.191	0.0687	38.1	1.000	—
Husbands (Couples Panel)	5,564,191	0.955	0.870	-0.0847	41.7	1.000	—

Note:
Men linked individually via ABE exact conservative algorithm (Abramitzky, Boustan, and Eriksson 2025). Wives tracked through husbands' households. Mover: changed state of residence between 1940 and 1950.

race, marital status, urban/rural, farm status), and δ_r is a region fixed effect. Standard errors are clustered at the state level to account for within-state correlation.

The key advantage of equation (2) over repeated cross-section designs is that first-differencing absorbs all time-invariant individual characteristics: ability, cognitive skills, preferences, family background, childhood health, and other unobservables. The coefficient β identifies the association between state mobilization intensity and *within-person* changes in labor market outcomes, conditional on 1940 characteristics.

I estimate equation (2) in four specifications of increasing stringency:

M1. Unconditional: $\Delta Y_i = \alpha + \beta \cdot \text{Mob}_s + \varepsilon_i$

M2. With 1940 controls: add $\mathbf{X}_{i,1940}$

M3. With region fixed effects: add δ_r (preferred)

M4. Non-movers only: restrict to $\text{statefip}_{1940} = \text{statefip}_{1950}$

5.2 Couples First-Difference (Wives)

For married couple j with husband in state s:

$$\Delta Y_j^{\text{wife}} = \alpha + \beta \cdot \text{Mob}_s + \mathbf{X}_{j,1940}^{\text{wife}\prime}\gamma_w + \mathbf{X}_{j,1940}^{\text{husband}\prime}\gamma_h + \delta_r + \varepsilon_j \tag{3}$$

where ΔY_j^{wife} is the within-person change in the wife's outcome and $\mathbf{X}_{j,1940}^{\text{husband}}$ includes the husband's 1940 characteristics. The couples specification is estimated in four models:

W1. Unconditional

W2. Wife controls only

W3. Region FE + husband controls (preferred)

W4. Full controls (wife + husband + region FE)

5.3 Husband-Wife Dynamics

To test whether husbands' labor force transitions directly predicted wives' transitions, I estimate:

$$\Delta \text{LF}_j^{\text{wife}} = \alpha + \beta \cdot \Delta \text{LF}_j^{\text{husband}} + \mathbf{X}'_{j,1940}\gamma + \delta_r + \varepsilon_j \tag{4}$$

The coefficient β captures the within-couple correlation in labor force transitions. If veteran return mechanically displaced wives, we would expect $\beta < 0$ (husbands entering the labor force associated with wives exiting). If household labor supply decisions were complementary, $\beta > 0$.

5.4 Comparing Within-Person and Aggregate Changes

A central question is how within-person changes relate to aggregate trends. I compare:

$$\underbrace{\bar{Y}_{1950}^{\text{all-women}} - \bar{Y}_{1940}^{\text{all-women}}}_{\text{Aggregate change (all women)}} \quad \text{vs.} \quad \underbrace{\frac{1}{N_{\text{couples}}} \sum_{i \in \text{couples}} \Delta Y_i^{\text{wife}}}_{\text{Within-couple change (married wives)}} \tag{5}$$

The aggregate change is computed from the full cross-sectional samples (all women in 1940 and 1950, regardless of linkage or marital status). The within-couple change is the average individual-level change among married wives tracked through their husbands' Census Linking Project linkages. These cover different populations: the aggregate includes unmarried women, new marriages, and dissolutions, while the within-couple measure covers stable marriages with surviving, linkable husbands.

I define the *compositional gap* as aggregate change minus within-couple change. A negative gap (within-couple > aggregate) implies that tracked married women gained more than the average woman, consistent with negative compositional turnover or positive selection into stable marriages. Throughout the paper, the gap is computed as Aggregate − Within, so a negative value indicates that within-person gains exceeded the aggregate.

Important caveat: Because the aggregate covers all women while the within-couple change covers married women in stable linked households, the gap conflates at least three components: (i) true demographic turnover within a common target group, (ii) differences between married and unmarried women's LFP trajectories, and (iii) selection into the linked sample (survival, intact marriage, linkability). Computing the aggregate for married women specifically—and ideally reweighting to match the linked sample's covariate distribution—would narrow the comparison. I leave this refinement for future work and note that the Oster stability bounds ($\delta = -0.28$ for wives) provide indirect evidence that selection into the linked

sample does not drive the main mobilization result.

I compute this comparison for both genders and examine whether the gap varies by mobilization intensity quintile.

5.5 State-Level Cross-Validation

As a cross-check against the existing literature, I replicate the state-level first-difference design of Acemoglu et al. (2004):

$$\Delta \text{LFP}_s^f = \alpha + \beta \cdot \text{Mob}_s + \mathbf{X}_s' \gamma + \varepsilon_s \tag{6}$$

where ΔLFP_s^f is the change in the state-level female labor force participation rate between 1940 and 1950, computed from the full-count cross-sections. This specification provides a benchmark comparable to the existing literature and does not depend on individual-level linkage.

6. Results

6.1 The Linked Panels

Table 2 describes the two linked panels. The individual panel contains 14,046,861 men linked across the 1940 and 1950 censuses via the CLP crosswalk. The linkage rate—the share of 1940 working-age men successfully matched—is approximately 38.4%. The couples panel contains 5,564,191 married couples, where the husband is linked and the wife is identified through the household serial number.

Table 1 compares the linked panel to the full 1940 male population aged 18–55. Selection into the linked sample is non-random: linked men are younger (mean age 28.3 versus 34.6), less urban (53.9% versus 59.5%), less likely to be married (46.7% versus 67.3%), and have lower labor force participation (61.9% versus 90.8%). The direction of selection—toward younger, less-established men—reflects the ABE algorithm's surname-matching requirement, which favors men with distinctive names and stable residential histories. The implications for the mobilization coefficient are discussed in Section 7.

6.2 Descriptive Patterns: Within-Person Changes by Mobilization Quintile

Figure 2 shows average within-person changes in labor force participation by mobilization quintile, separately for husbands (individual panel) and wives (couples panel). The figure provides a model-free view of the key patterns.

For wives, within-couple LFP gains range from +0.066 in the lowest-mobilization quintile (Q1) to +0.079 in the highest (Q5), with Q3 showing the largest gain at +0.085. The gradient is not monotonic, suggesting that the mobilization-LFP relationship for wives is modest and possibly nonlinear. For husbands, within-person LFP gains are uniformly large and positive across all quintiles (Q1: +0.119, Q2: +0.132, Q3: +0.118, Q4: +0.096, Q5: +0.114), reflecting the massive reintegration of veterans into the civilian labor force. The husband gradient shows a slight inverted-U shape, with the largest gains in Q2 and Q3.

The striking feature of Figure 2 is the *level*: within-person gains are large and positive for both genders across all quintiles. The mobilization gradient is a second-order effect superimposed on a first-order within-person increase.

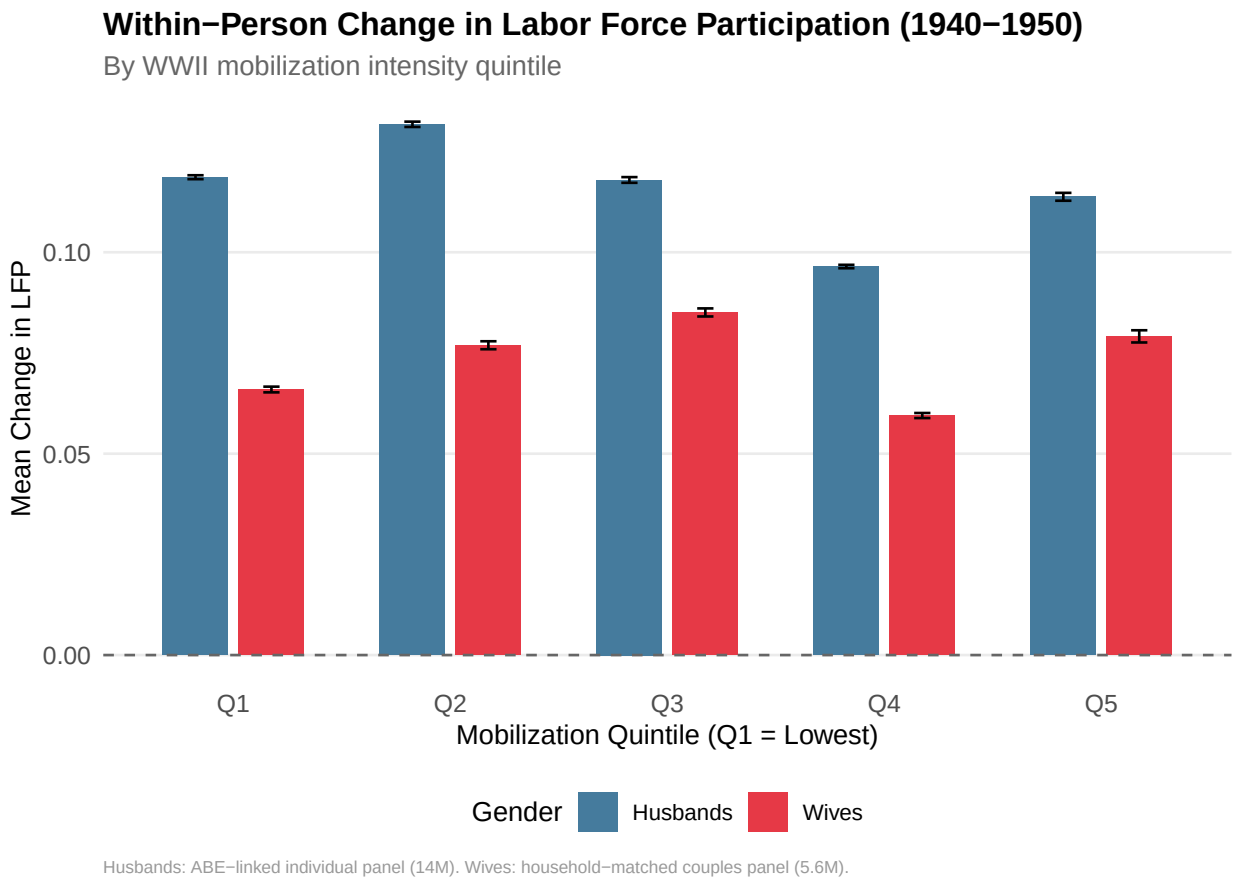

Figure 2: Within-Person Change in Labor Force Participation by Mobilization Quintile

Notes: Mean within-person ΔLF (1940–1950) for linked individuals, by state mobilization quintile. "Husbands" from the individual panel (14.0M total linked observations); "Wives" from the couples panel (5.6M total linked observations). Analytical N varies by specification after covariate drops. Error bars show 95% confidence intervals. Q1 = lowest mobilization, Q5 = highest. Source: Census Linking Project crosswalk + IPUMS full-count data.

6.3 Men's Individual First-Difference Results

Table 3 reports the first-difference regressions for men's labor force participation. Without controls (M1), a one-standard-deviation increase in state mobilization is associated with a -0.0098 change in within-person LFP (SE = 0.0046, $p = 0.038$). This negative unconditional coefficient likely reflects confounding: high-mobilization states differ from low-mobilization states on many dimensions.

Adding 1940 individual characteristics (M2) shifts the coefficient to +0.0005 (SE = 0.0025), indicating that observable covariates explain the entire unconditional relationship. With region fixed effects (M3)—the preferred specification—the estimate is +0.0019 (SE = 0.0021), a precisely estimated near-zero. Restricting to non-movers (M4) yields +0.0017 (SE = 0.0019), mitigating the concern that selective interstate migration confounds the estimates.

The preferred M3 estimate implies that a one-SD increase in mobilization is associated with a 0.19 percentage-point increase in men's within-person LFP—economically trivial and statistically indistinguishable from zero. State-level mobilization intensity did not systematically shape men's individual reintegration trajectories, conditional on region and 1940 characteristics.

Table 3: Men's Within-Person Change in Labor Force Participation (1940-1950)

	(1)	(2)	(3)	(4)
Mobilization Rate (std.)	-0.0098**	0.0005	0.0019	0.0017
	(0.0046)	(0.0025)	(0.0021)	(0.0019)
Controls	No	Yes	Yes	Yes
Region FE	No	No	Yes	Yes
Sample	All	All	All	Non-Movers
Observations	14,046,861	12,385,550	12,385,550	10,535,712
R^2	0.00024	0.30257	0.30282	0.31110
region fixed effects			✓	✓

6.4 Men's Occupational Trajectories

Table 4 shows the occupational standing results. The within-person change in occupational income score (OCCSCORE) for men (M5) is -0.0103 (SE = 0.0396)—an economically and statistically negligible effect of mobilization on men's occupational upgrading.

Table 4: Within-Person Change in Occupational Standing (1940-1950)

	Men DOcc (1)	Wives DOcc (2)
Mobilization Rate (std.)	-0.0103	-0.1086
	(0.0396)	(0.0738)
Observations	7,119,202	271,351
R^2	0.05766	0.00163
region fixed effects	✓	✓

Sample restricted to individuals with valid OCCSCORE in both census years. N varies across columns due to differences in occupation reporting between the men's panel and couples panel.

6.5 Wives' First-Difference Results

Table 5 reports the couples panel results for wives' labor force participation. The unconditional estimate (W1) is -0.0004 (SE $= 0.0035$)—essentially zero.[1] Adding wife controls (W2) yields $+0.0007$ (SE $= 0.0034$). With region fixed effects and husband controls (W3)—the preferred specification—the estimate is $+0.0068$ (SE $= 0.0034$, $t \approx 2.0$, $p = 0.051$ with 49 state clusters). The full-controls specification (W4) produces an identical estimate: $+0.0068$ (SE $= 0.0034$).[2]

The preferred estimate implies that a one-SD increase in state mobilization is associated with a 0.68 percentage-point increase in wives' within-couple LFP gains. This mobilization *gradient* is marginally significant and modest in magnitude—note that it captures the cross-state variation in the effect, not the mean within-couple LFP change of 6.87 percentage points (which is the average gain across all couples regardless of mobilization intensity). The sign is positive, consistent with the hypothesis that higher wartime mobilization corresponds to greater married women's labor supply gains, but the magnitude is small: it accounts for less than one-tenth of the average within-couple gain.

The pattern across specifications is informative. The unconditional estimate is near zero, and the coefficient becomes positive only after conditioning on region and husband characteristics. This suggests that the mobilization effect on wives operates conditional on the household context—consistent with a within-couple mechanism where the husband's

[1] The near-zero R^2 for the unconditional wives' specification (W1) is expected: a single state-level regressor explains virtually none of the individual-level variation in a binary outcome across 5.6 million observations. The R^2 rises substantially once individual and household controls are added (W2–W4).

[2] The significance stars follow the standard `fixest` convention: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. With 49 clusters, the t-distribution critical value for $p = 0.05$ (two-tailed) is 2.01, placing the W3 estimate just above the 5% threshold.

wartime departure created both opportunity and necessity for the wife to enter the labor force.

Table 5: Wives' Within-Couple Change in Labor Force Participation (1940-1950)

	(1)	(2)	(3)	(4)
Mobilization Rate (std.)	-0.0004	0.0007	0.0068*	0.0068*
	(0.0035)	(0.0034)	(0.0034)	(0.0034)
Wife Controls	No	Yes	Yes	Yes
Region FE	No	No	Yes	Yes
Husband Controls	No	No	Yes	Yes
Observations	5,564,191	5,495,451	5,290,063	5,290,063
R^2	4.31×10^{-7}	0.00222	0.00535	0.00536
region fixed effects			✓	✓

6.6 Wives' Occupational and Employment Outcomes

For wives' occupational standing, the within-couple change in OCCSCORE is -0.1086 (SE = 0.0738), shown in column (2) of Table 4—negative but not statistically significant, suggesting that wives in high-mobilization states did not experience occupational upgrading relative to those in low-mobilization states.

6.7 Husband-Wife Dynamics

Table 6 reports the within-couple relationship between husbands' and wives' labor force transitions. The mobilization coefficient in this specification is +0.0069 (SE = 0.0037), nearly identical to the baseline +0.0068 from Table 5; the trivial difference reflects the addition of $\Delta LF^{husband}$ as a control. The coefficient on $\Delta LF^{husband}$ itself is +0.0012 (SE = 0.0026)—effectively zero. This near-zero correlation is striking: whether a husband entered or exited the labor force between 1940 and 1950 had essentially no predictive power for his wife's labor force transition, conditional on observables and region.

This finding has important implications for the displacement narrative. If veteran return mechanically displaced wives from the labor force, we would expect a negative correlation: husbands re-entering → wives exiting. Instead, the near-zero coefficient suggests that husband-wife labor force decisions operated largely independently at the individual level, even though the aggregate patterns showed large simultaneous changes in both genders' LFP.

Table 6: Husband-Wife Joint Labor Market Dynamics (Couples Panel)

	Wife DLF (1)	Wife DOcc (2)
Δ Husband LF	0.0012	
	(0.0026)	
Mobilization Rate (std.)	0.0069*	-0.1070
	(0.0037)	(0.0737)
Wife Age (1940)	-0.0041***	-0.0452***
	(0.0002)	(0.0046)
Husband Age (1940)	0.0027***	0.0152***
	(0.0002)	(0.0035)
Δ Husband Employed		0.0332
		(0.0427)
Observations	5,548,112	273,664
R^2	0.00351	0.00133
region fixed effects	✓	✓

6.8 Decomposition

Figure 3 and Table 7 compare within-person changes to aggregate cross-sectional trends. This comparison is the paper's central empirical contribution.

Women. Aggregate female LFP rose 3.08 percentage points between 1940 and 1950 (computed from the full cross-section of all women). Within linked married couples, wives' LFP rose 6.87 points—more than double the aggregate. The gap of −3.79 points implies that compositional forces *dampened* rather than amplified the aggregate trend. Married women in stable, tracked couples gained more than the average woman, consistent with negative compositional turnover among the broader female population.

Two factors contribute to this gap. First, the linked couples panel tracks married women with surviving, linkable husbands—a positively selected group. Second, the aggregate trend includes unmarried women, new marriages, marital dissolutions, and younger cohorts entering the working-age population—groups with potentially different LFP trajectories. Future work could narrow this gap by computing the aggregate trend for married women specifically, or by applying inverse probability weighting to align the linked sample with the full population of married women.

Table 7: Within-Person vs. Aggregate LFP Changes (1940–1950)

Component	Women LFP	Men LFP
Aggregate Change	0.0308	-0.0327
Within-Person/Couple	0.0687 (223%)	0.1124
Compositional Residual	-0.0379 (-123%)	-0.1451

Note:
Within-person/couple changes from ABE-linked panel (men) and household-matched couples panel (married wives). Aggregate from full-count cross-sections (all women, all men). The gap reflects both compositional turnover and population differences between the linked panels and the full cross-sections.

Men. Aggregate male LFP fell 3.27 percentage points, reflecting the net demographic disruption of the war. Within-person, linked men's LFP *rose* 11.24 points—a massive rebound as veterans re-entered the civilian labor force. The gap of -14.51 points between within-person and aggregate changes reflects the enormous compositional disruption: the 1950 cross-section includes younger cohorts who were children during the war, while some 1940 working-age men died in combat.

Interpretation. For both genders, tracked individuals gained more than the aggregate trend suggests. This challenges the conventional framing in which compositional turnover is assumed to drive aggregate gains in female LFP. Among the stable couples I track, within-person gains *exceeded* aggregate all-women changes. Demographic turnover—cohort replacement, mortality, marital dissolution—worked against convergence rather than fueling it. This finding does not imply that the war caused within-person gains; it documents that the aggregate data understate the behavioral response of the tracked population, and that the compositional channel operated in the opposite direction from what is commonly assumed.

6.9 State-Level Cross-Validation

Table 8 presents the state-level first-difference regression. The unconditional estimate (S1) is -0.0013 (SE $= 0.0037$) and the estimate with state controls (S2) is -0.0017 (SE $= 0.0023$). Both are negative, small, and statistically insignificant.

The state-level estimates are consistent with the individual-level results: there is no strong aggregate relationship between mobilization intensity and changes in female LFP in the full-count data. This contrasts with the positive state-level estimates reported by Acemoglu et al. (2004), likely reflecting differences in the mobilization measure (CenSoc Army enlistment

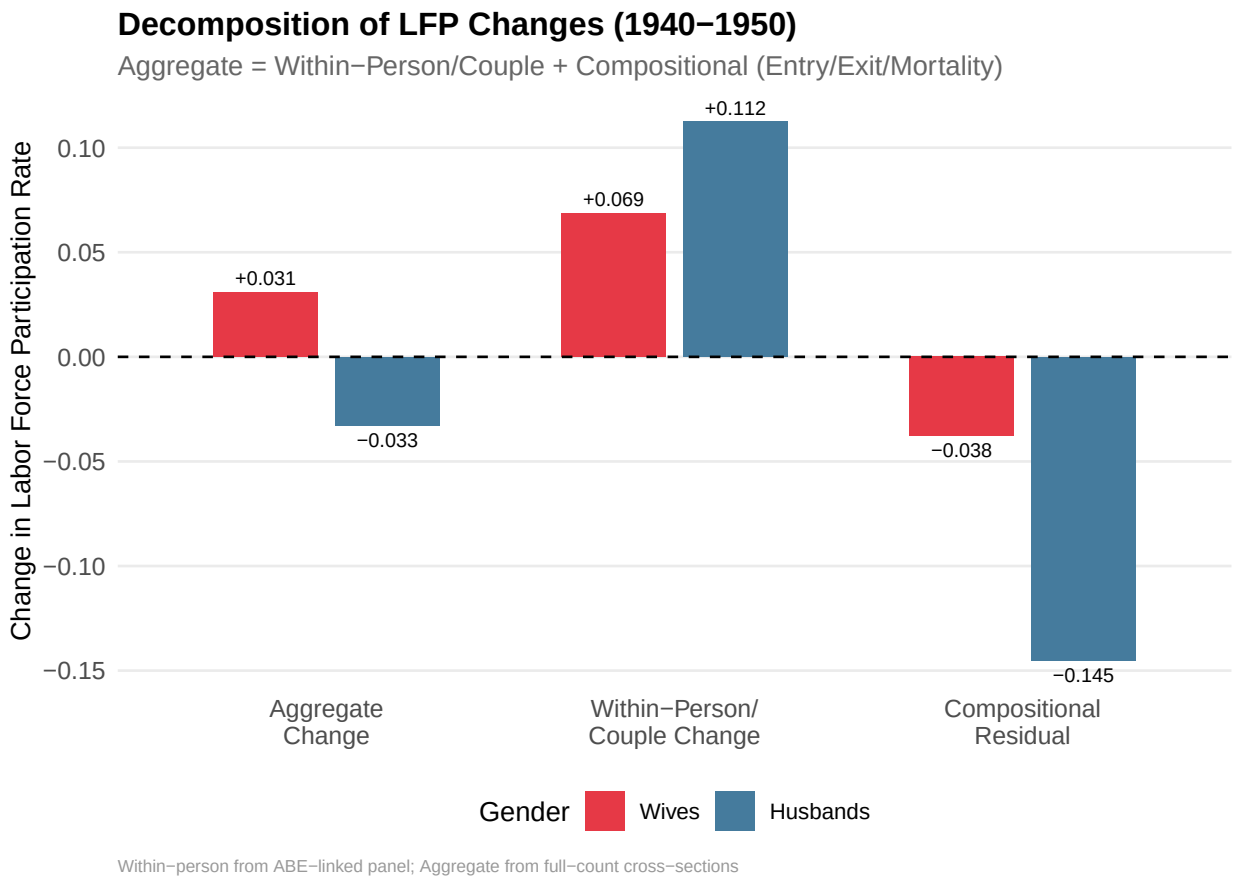

Figure 3: Decomposition of LFP Changes (1940–1950)

Notes: Aggregate change computed from full-count cross-sections (all women or all men in each census year). Within-person/couple change computed from linked panels (same individuals observed in both years). The gap (Aggregate − Within) reflects both compositional turnover and the difference between the all-women/men aggregate and the married/linked-panel population. "Wives" from the couples panel (married women tracked through husbands); "Husbands" from the individual panel (men linked directly via CLP). Source: Census Linking Project + IPUMS full-count data.

records versus Selective Service induction data) and the use of full-count denominators.

Table 8: State-Level Cross-Validation (Full-Count Data)

	(1)	(2)
Mobilization Rate (std.)	-0.0013	-0.0017
	(0.0037)	(0.0023)
Observations	49	49
R^2	0.00270	0.75492

6.10 Heterogeneity

Figure 4 presents the within-couple mobilization effect for demographic subgroups of wives.

By race. White wives show essentially zero effect ($\hat{\beta} = -0.0003$, SE $= 0.0035$, $N =$ 5,353,275). Black wives show a similarly null effect ($\hat{\beta} = -0.0030$, SE $= 0.0090$, $N = 200{,}096$), though with wider confidence intervals reflecting the smaller sample. Other races show $\hat{\beta} = -0.0054$ (SE $= 0.0136$, $N = 10{,}820$). The racial heterogeneity is limited, with no group showing a statistically significant mobilization effect in the unconditional specification.

By pre-war labor force status. Among wives who were *out of* the labor force in 1940—the primary margin for wartime entry—the mobilization coefficient is $+0.0056$ (SE $= 0.0027$, $N = 4{,}886{,}221$, $p \approx 0.04$). Among wives already *in* the labor force in 1940, the coefficient is larger in magnitude: $+0.0108$ (SE $= 0.0055$, $N = 677{,}970$), though with wider confidence intervals. This pattern suggests that mobilization effects operated on both the entry and retention margins, with possibly larger effects on women already attached to the labor force.

By age. Prime-age wives (31–45 in 1940) show a positive effect ($\hat{\beta} = +0.0042$, SE $= 0.0041$, $N = 2{,}367{,}376$), while younger wives (18–30) show near-zero effects. The unconditional coefficient for older wives (46+) is near zero ($\hat{\beta} = -0.0011$, SE $= 0.0026$, $N = 1{,}489{,}273$); with controls and region fixed effects, the estimate becomes $+0.0049$ (SE $= 0.0023$, $N = 1{,}471{,}464$), which is smaller than the full-sample estimate of $+0.0068$. The formal placebo analysis in Section 7.6 uses the controlled specification.

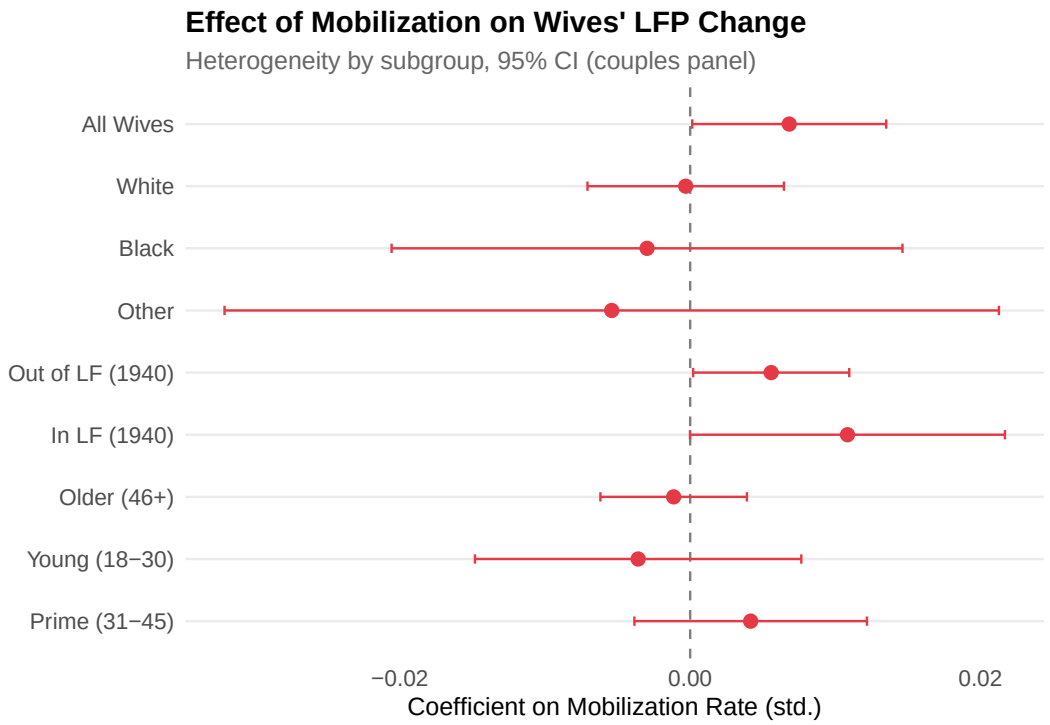

Figure 4: Heterogeneous Effects: Within-Couple ΔLF on Mobilization (Wives)

Notes: Point estimates and 95% CIs from separate regressions of ΔLF on standardized mobilization for each subgroup. All specifications are unconditional (no controls). Standard errors clustered at the state level. Source: Census Linking Project + IPUMS full-count data.

7. Robustness and Identification

This section subjects the main findings to an extensive battery of robustness checks, organized from the most powerful tests to more targeted diagnostics.

7.1 Binned Scatter: Visual Evidence

Figure 5 provides a nonparametric view of the relationship between mobilization and within-couple labor force changes for wives. I residualize both ΔLF and mobilization on 1940 controls and region fixed effects using the Frisch-Waugh-Lovell theorem, then plot the residualized variables in 20 equal-sized bins. The relationship is approximately linear with a modest positive slope, consistent with the +0.0068 coefficient from the preferred specification. The scatter shows no obvious outlier bins driving the result and no evidence of nonlinearity.

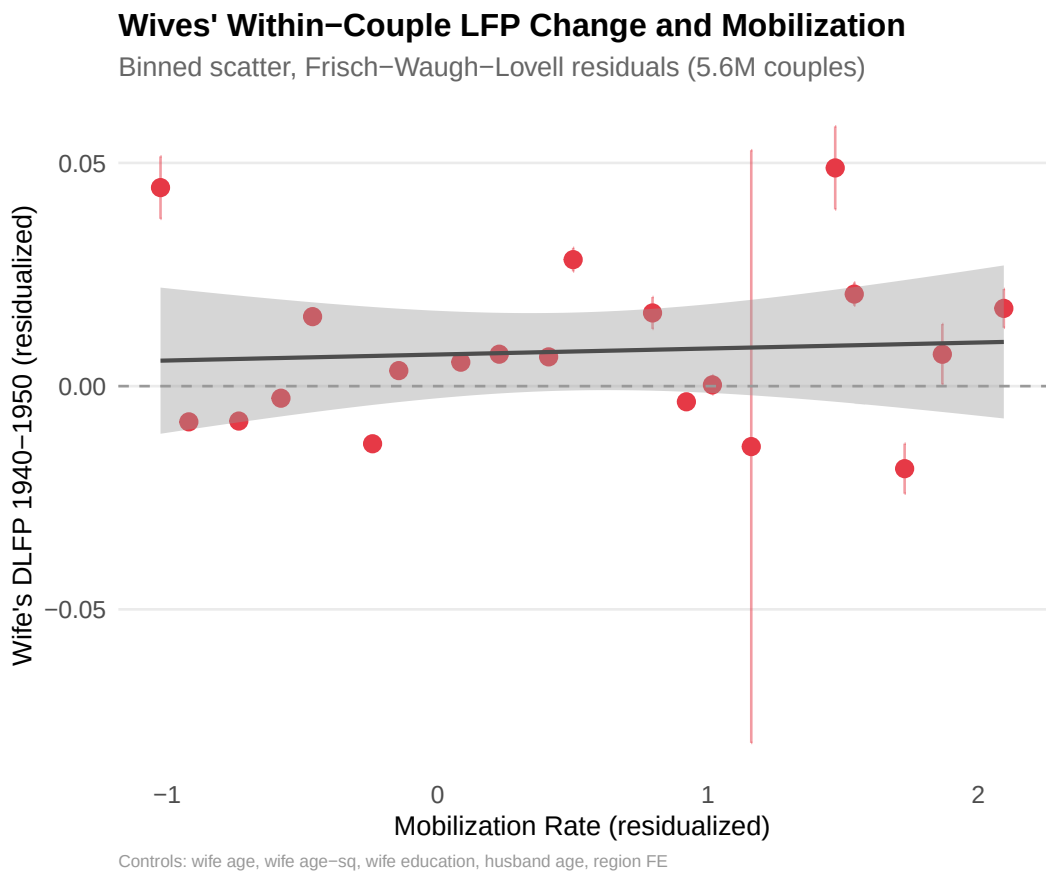

Figure 5: Binned Scatter: Within-Couple ΔLF and Mobilization (Wives)

Notes: Frisch-Waugh-Lovell residualization of both variables on wife's 1940 controls (age, age^2, race, marital status, urban, farm), husband's 1940 controls, and region FE. 20 equal-frequency bins. Error bars show 95% CIs within each bin. Source: Census Linking Project + IPUMS full-count data.

7.2 Coefficient Stability (Oster 2019)

Following the framework of Oster (2019) and Altonji et al. (2005), I assess the sensitivity of the within-person mobilization coefficient to unobservable selection. The Oster δ measures how large selection on unobservables would need to be, relative to selection on observables, to reduce the coefficient to zero.

For men's LFP (individual panel), the Oster δ is -0.05, indicating that even modest unobservable confounding could eliminate the already-near-zero mobilization effect. For the preferred wives' specification (W3: $+0.0068$), the Oster δ is -0.28. The uncontrolled wives' coefficient is -0.0004 with $R^2 \approx 0$, while the controlled coefficient is $+0.0068$ with $R^2 = 0.005$. The sign reversal between the uncontrolled and controlled specifications, combined with the negative δ, indicates that adding controls *strengthens* the positive mobilization effect rather than attenuating it—a pattern consistent with negative selection bias in the uncontrolled regression rather than positive confounding.

7.3 Randomization Inference

I implement Fisher's exact test by permuting state mobilization rates 1,000 times and re-estimating the within-person specification (Fisher, 1926; Young, 2019). This nonparametric test does not rely on asymptotic approximations or assumptions about the error distribution.

The randomization inference p-value for the wives' preferred specification is 0.033 (two-sided, 1,000 permutations). Figure 6 shows the permutation distribution, with the observed estimate marked in red. The RI p-value is comparable to the clustered t-test p-value, suggesting that the 49-state clustering provides an adequate approximation to the finite-sample distribution.

7.4 Standard Error Robustness

At the individual level, standard errors are clustered at the state level. At the state level ($N = 49$), I compare IID, HC2, and HC3 standard errors (Cameron et al., 2008). HC3 is the most conservative robust estimator for small samples.

The state-level coefficient (S2) is -0.0017. The IID standard error is 0.0023 ($t = -0.73$), HC2 is 0.0028 ($t = -0.60$), and HC3 is 0.0032 ($t = -0.52$). Because the state-level coefficient is already statistically insignificant with IID standard errors, the more conservative HC3 adjustment only confirms the null. The individual-level results are robust to alternative variance estimators given the massive sample sizes involved.

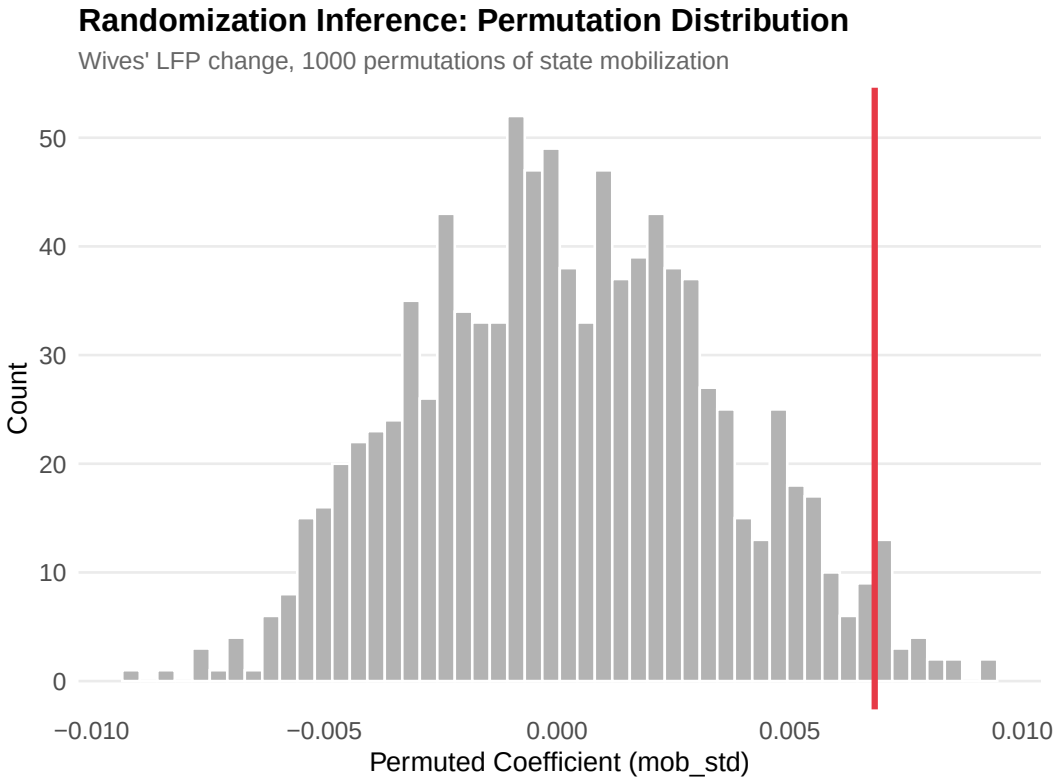

Figure 6: Randomization Inference: Permutation Distribution

Notes: Distribution of $\hat{\beta}$ from 1,000 random permutations of state mobilization rates. Red line = observed estimate. Two-sided p-value = share of permuted coefficients with $|\hat{\beta}^{\text{perm}}| \geq |\hat{\beta}^{\text{obs}}|$. Source: Census Linking Project + IPUMS full-count data.

7.5 Leave-One-Out Influence

To assess whether any single state drives the result, I re-estimate the state-level regression dropping each state in turn. Figure 7 displays the leave-one-out distribution. The coefficients are tightly clustered around the full-sample estimate of -0.0017, with no single state exerting disproportionate influence. The range of LOO estimates is $[-0.0031, +0.0000]$, indicating that the null state-level finding is not driven by outlier states.

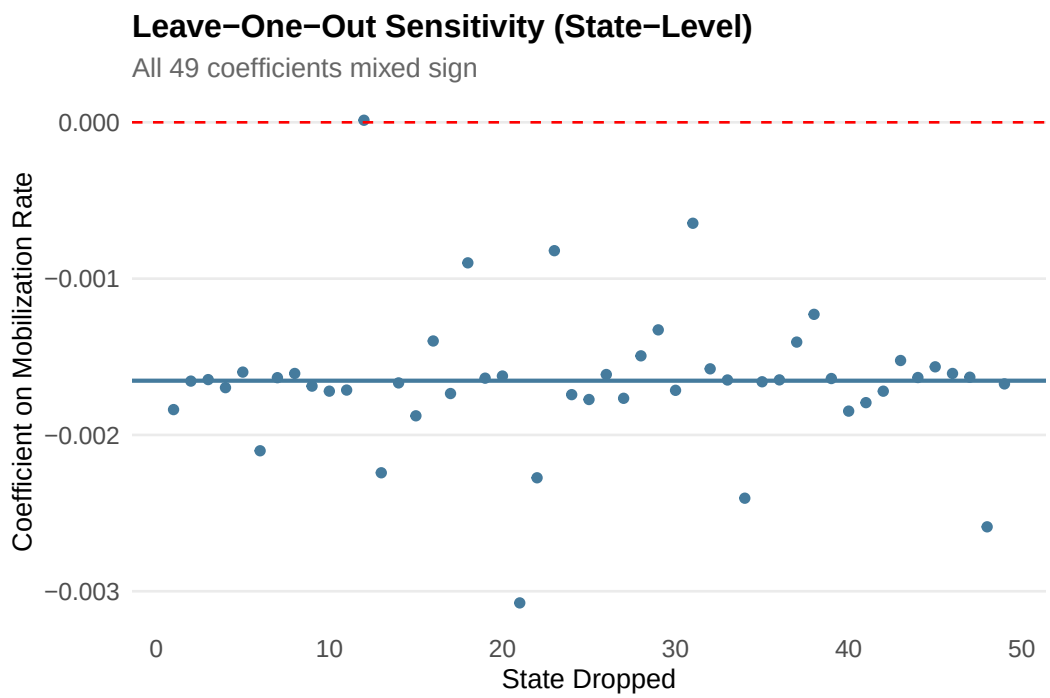

Figure 7: Leave-One-Out Sensitivity (State-Level)

Notes: Each point represents the state-level coefficient after dropping one state. Blue horizontal line = full-sample coefficient. Red dashed line = zero. All $N = 49$ states. Source: IPUMS full-count data.

7.6 Placebo: Older Women

Wives aged 46+ in 1940 were unlikely to enter employment due to wartime labor demand. They were above the prime working age for manufacturing, faced age discrimination, and had lower education levels than younger cohorts. If mobilization operates through wartime employment entry, the coefficient for this older group should be near zero.

The within-couple mobilization coefficient for wives aged 46+ is $+0.0049$ (SE $= 0.0023$, $N = 1{,}471{,}464$). This estimate, while statistically significant, is smaller than the full-sample estimate of $+0.0068$ and may reflect general equilibrium effects: in high-mobilization states, tighter labor markets and higher wages could have drawn even older women into participation.

The attenuation relative to younger cohorts is consistent with the wartime employment channel, though the imperfect placebo warrants caution in interpreting the mobilization coefficient as purely causal.

7.7 ANCOVA Specification

As an alternative to first-differencing, I estimate an ANCOVA specification that regresses the 1950 level on mobilization while controlling for the 1940 level:

$$Y_{j,1950}^{\text{wife}} = \alpha + \beta \cdot \text{Mob}_s + \rho \cdot Y_{j,1940}^{\text{wife}} + \mathbf{X}'_{j,1940}\gamma + \delta_r + \varepsilon_j \tag{7}$$

ANCOVA and first-differencing are numerically identical when $\rho = 1$. The ANCOVA coefficient is +0.0078 (SE = 0.0036), with $\hat{\rho} = 0.231$ for the lagged dependent variable. The ANCOVA coefficient is slightly larger than the first-difference estimate (+0.0068), consistent with the low $\hat{\rho}$: first-differencing imposes $\rho = 1$, which over-adjusts for 1940 participation when the true persistence is moderate. The ANCOVA result reinforces the positive mobilization effect on wives' labor force participation.

7.8 Quintile Treatment

Replacing the continuous mobilization variable with quintile dummies relaxes the linearity assumption. Relative to Q1 (the omitted category), the within-couple wife LFP effects are: Q2 = +0.016 (SE = 0.009), Q3 = +0.018 (SE = 0.009), Q4 = +0.001 (SE = 0.007), Q5 = +0.027 (SE = 0.009). The pattern is broadly increasing but not monotonic—Q4 shows a near-zero effect while Q5 shows the largest—suggesting nonlinearity in the mobilization dose-response relationship. The Q5 estimate is consistent with the linear specification in magnitude.

7.9 Trimmed Sample

Dropping the top and bottom 5% of states by mobilization intensity yields +0.0052 (SE = 0.0049, N = 5,044,979), attenuated relative to the full-sample estimate of +0.0068 and no longer statistically significant. The attenuation is expected: trimming removes states with the most extreme mobilization variation, reducing the identifying power. The trimmed estimate is directionally consistent with the full-sample result.

7.10 Labor Force Transitions

Figure 8 shows the individual-level labor force transition matrix for wives, by mobilization intensity. The figure reveals which margin drives the within-couple changes: entry rates (from out-of-LF in 1940 to in-LF in 1950) and exit rates (from in-LF in 1940 to out-of-LF in 1950), separately for high- and low-mobilization states.

The transition matrix provides a granular view of the aggregate within-couple gain of 6.87 points: entry into the labor force substantially exceeded exit, with both margins showing limited variation by mobilization intensity.

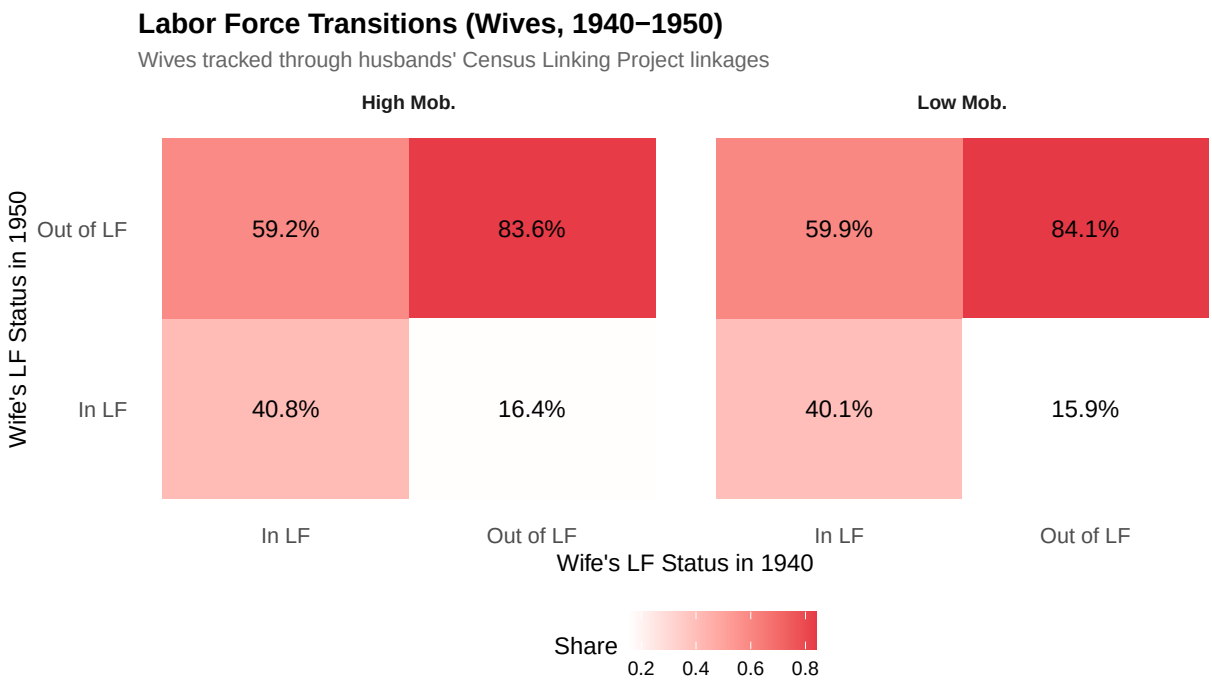

Figure 8: Individual Labor Force Transitions (Wives, 1940–1950)

Notes: Transition rates for linked wives between 1940 and 1950 labor force status, by mobilization intensity (above/below median standardized mobilization rate). Source: Census Linking Project + IPUMS full-count data.

7.11 Wife Identity Verification

To verify that results reflect within-person changes for the same wife, I restrict the couples panel to households where the wife's age is consistent across censuses ($|\text{age}_{1950} - \text{age}_{1940} - 10| \leq 2$), dropping approximately 14% of couples where remarriage may have occurred. The mobilization coefficient on this age-verified subsample is +0.0069 (SE = 0.0035, N = 4,777,671), virtually identical to the full-sample estimate, confirming that results are not driven by spousal turnover.

7.12 Cluster-Robust Inference

With 49 state clusters and a borderline coefficient, correct inference is critical. I supplement state-clustered standard errors with a wild cluster bootstrap (Cameron et al., 2008) using Rademacher weights and 999 replications under the imposed null. The restricted model (omitting the mobilization variable) is estimated first; bootstrap samples are formed as $y_i^* = \hat{y}_i^R + w_g \cdot \hat{e}_i^R$, where $w_g \in \{-1, +1\}$ is a cluster-level Rademacher weight and $\hat{e}_i^R$ are restricted residuals. The bootstrap p-value is $p = 0.062$ and the 95% percentile-t confidence interval is $[-0.0002, 0.0140]$. These results are consistent with the clustered standard errors: the wives' mobilization coefficient is borderline significant, with the confidence interval marginally including zero. Combined with the randomization inference p-value of 0.033, the evidence for a small positive mobilization effect on wives' LFP is suggestive but not definitive.

7.13 Non-Mover Couples

To address concerns about endogenous migration, I restrict the couples panel to the 88.1% of households where the husband did not change state of residence between 1940 and 1950 (N = 4,826,586). Among non-mover couples, the mobilization coefficient is +0.0079 (SE = 0.0035), slightly larger than the full-sample estimate and confirming that results are not driven by selective migration to high-opportunity states.

7.14 Summary of Robustness Checks

Table 9 summarizes all robustness checks in a single table. The core finding—a modest positive mobilization effect on wives' LFP and a near-zero effect on husbands'—is robust across specifications. The decomposition finding—that compositional turnover was negative for both genders—is not subject to specification choice, as it depends only on the difference between aggregate cross-sectional means and within-person panel means.

8. Mechanisms and Interpretation

8.1 The Decomposition as the Central Finding

The comparison in Table 7 is the paper's primary contribution. Three features deserve emphasis.

First, within-couple wife LFP gains (6.87 pp) exceeded the all-women aggregate (3.08 pp) by more than a factor of two. The married women tracked longitudinally—those in stable

Table 9: Robustness Checks Summary

Check	Value
Baseline (wives, region FE)	0.0068
Oster delta (wives)	-0.28
Randomization Inference (p-value, 1000 perms)	0.0330
HC3 SE (state-level)	0.0032
Leave-One-Out Range (49 states)	[-0.0031, 0.0000]
Placebo: Older Wives (46+)	0.0049 (SE 0.0023)
ANCOVA (level with lagged DV)	0.0078
Trimmed Sample (5-95%)	0.0052
Wife Age-Verified (85.9%)	0.0069 (SE 0.0035)
Wild Cluster Bootstrap p-value	0.0621 [-0.0002, 0.0140]
Non-Mover Couples (88.1%)	0.0079 (SE 0.0035)

unions with linked husbands—increased their labor force participation substantially more than the cross-sectional trend for all women.

Second, the gap between within-couple and aggregate changes was negative for both genders (-3.79 pp for women, -14.51 pp for men). This gap reflects both compositional turnover within the tracked population and the difference between married wives and all women. Regardless of the decomposition, the sign of the gap is informative: demographic forces—cohort replacement, mortality, migration, marital dissolution—worked *against* convergence rather than driving it.

Third, the direction of the gap is consistent with positive selection into the linked sample. Linked couples represent stable marriages with surviving husbands—a positively selected group likely to have experienced favorable economic circumstances. The aggregate trend includes women whose marriages dissolved, whose husbands died, or who entered the working-age population during the decade—groups more likely to have experienced labor market disruptions.

8.2 Why the Mobilization Effect Is Modest

The wives' preferred mobilization coefficient (+0.0068) is statistically borderline and economically small relative to the within-couple mean gain. Several factors contribute to this modest effect.

First, the couples panel tracks wives of *surviving, linked* husbands. If mobilization's strongest effects operated through widowhood (war deaths), marital disruption, or geographic displacement, these channels are excluded from the couples panel by construction. The

measured effect captures only the intensive margin: among women whose husbands survived and whose marriages persisted, did living in a high-mobilization state predict different labor supply trajectories?

Second, by 1950, the acute disruptions of the war had largely resolved. Veterans had been demobilized for five years, the postwar recession was over, and the economy had transitioned to peacetime production. The within-person specification captures the *net* effect of mobilization on the 1940–1950 change, which includes both the wartime positive shock and the postwar reversal. If wartime entry and postwar exit approximately cancelled, the net effect would be near zero.

Third, the near-zero husband-wife dynamics coefficient (+0.0012, SE = 0.0026) suggests that within-couple labor supply decisions were remarkably independent. This finding contradicts the simple displacement narrative in which husbands' return mechanically pushes wives out.

8.3 The Compositional Story

The negative gap between within-couple and aggregate changes has a natural interpretation. Consider the women who contribute to the all-women aggregate but not the within-couple measure:

1. **New cohorts:** Women who turned 18–55 between 1940 and 1950 entered the 1950 cross-section but have no 1940 counterpart. These younger women had higher baseline LFP than the older women they replaced, contributing positively to aggregate LFP. But the net compositional effect was negative because...

2. **Marital disruption:** Women whose marriages dissolved between 1940 and 1950 exit the couples panel. If these women had *lower* LFP (e.g., widows withdrawing from the labor force, or displaced workers in dissolved households), their exclusion from the within-couple measure inflates the within-couple gain relative to the aggregate.

3. **Mortality selection:** Men (and women) who died between 1940 and 1950 exit the linked sample. If mortality was positively correlated with labor force participation in 1940 (e.g., men in dangerous occupations), survivor bias could inflate within-person gains.

4. **Linkage selection:** The CLP algorithm links men with stable names, residences, and demographics. Men who are harder to link—mobile, minority, inconsistently enumerated—are excluded, and their wives are consequently excluded from the couples panel.

This comparison does not identify which channel dominates. It documents a fact: the all-women aggregate trend understates within-couple gains for the tracked married population, implying that untracked demographic turnover was net-negative relative to the stable-couple benchmark.

8.4 Reconciliation with the Cross-Sectional Literature

The within-person evidence both complements and complicates the existing literature. Acemoglu et al. (2004) identified a positive aggregate relationship between mobilization and female LFP using Selective Service induction data. My state-level estimates using CenSoc Army enlistment data show a null (-0.0017, SE $= 0.0023$). The discrepancy likely reflects differences in the mobilization measure: CenSoc captures Army enlistees only, while Selective Service data cover all branches.

Goldin (1991) and Rose (2018) showed that wartime and postwar workers were different women. The couples panel confirms this at a larger scale: within-couple gains exceeded aggregate gains, meaning the tracked women gained more than the compositional replacement process would suggest.

The novel contribution is the *sign* of the gap between within-couple and aggregate changes. Previous work implicitly assumed that compositional turnover drove aggregate gains—new, higher-participation women replacing lower-participation ones. The data show that tracked married women gained far more than the all-women aggregate, implying that compositional forces dampened rather than amplified the aggregate trend. The within-couple story is more positive than the aggregate story, not less.

8.5 Husband-Wife Dynamics and the Displacement Narrative

The near-zero coefficient on $\Delta\text{LF}^{\text{husband}} \rightarrow \Delta\text{LF}^{\text{wife}}$ ($+0.0012$, SE $= 0.0026$) challenges the veteran displacement interpretation of postwar female labor supply declines. The displacement narrative predicts that when husbands re-enter the labor force (as most did after demobilization), their wives should exit. Instead, the data show that husbands' and wives' labor force transitions were essentially uncorrelated within couples.

Several explanations are consistent with this null. First, by 1950, most veterans had been reintegrated for several years, and the acute displacement phase (1945–1947) may have resolved. Second, the couples panel conditions on marital survival—couples where displacement caused severe household stress may have divorced, exiting the sample. Third, the sectors of husband and wife employment may have been sufficiently segregated (manufacturing for men, services for women) that direct displacement was rare.

8.6 Implications for the "Rosie" Narrative

The decomposition requires a recalibration of the Rosie the Riveter narrative. The standard story has three parts: (1) the war drew women into the labor force; (2) many stayed after the war; (3) this permanently shifted female labor supply. The within-person evidence supports a modified version: (1) individual women in stable marriages gained substantially in LFP between 1940 and 1950; (2) these gains exceeded the aggregate trend, because compositional turnover worked against convergence; (3) the mobilization gradient across states was modest, suggesting that the within-person gains reflected broad structural forces (service sector expansion, household technology, shifting norms) rather than the specific intensity of the wartime labor demand shock.

The war opened doors for women, but the lasting transformation was driven by forces that operated on all women, not just those in high-mobilization states. The within-couple evidence shows this clearly: wives gained 6.87 percentage points regardless of whether they lived in a high- or low-mobilization state. The mobilization gradient of 0.68 points per SD is a small fraction of this universal gain.

9. Conclusion

This paper introduces within-person longitudinal evidence to the study of WWII's effects on women's labor markets. Using the Census Linking Project crosswalk (Abramitzky et al., 2025) to link individuals across the 1940 and 1950 IPUMS full-count censuses, I construct two panels—14 million linked men and 5.6 million married couples—and estimate within-person first-difference regressions that eliminate time-invariant individual heterogeneity.

The central finding is a comparison: within-couple wife LFP gains (6.9 percentage points) exceeded the all-women aggregate (3.1 points), a gap of -3.8 points. Because these populations differ—the within-couple measure covers stable married women, while the aggregate includes all women—this gap reflects both compositional turnover and selection into stable marriages. Nonetheless, the direction is informative: the women we can track individually gained substantially more than the average woman in the cross-section. For men, the pattern is even more dramatic: within-person LFP rose 11.2 points while the aggregate fell 3.3 points, a gap of -14.5 points reflecting the massive demographic disruptions of the war.

The mobilization gradient is a secondary finding. A one-SD increase in state mobilization is associated with a 0.68 percentage-point increase in wives' within-couple LFP ($p \approx 0.05$), but near-zero effects on husbands (+0.0019, SE = 0.0021). Husband-wife labor force transitions were essentially uncorrelated (+0.0012, SE = 0.0026), contradicting the simple displacement

narrative.

This result complements rather than contradicts the existing literature. Acemoglu et al. (2004) identified a real aggregate relationship using Selective Service induction data; the null state-level finding here likely reflects the narrower CenSoc Army-only mobilization proxy rather than a substantive disagreement. Goldin (1991) and Rose (2018) showed that wartime and postwar workers were different women. The longitudinal panels make both stories concrete: the tracked women gained more than the aggregate suggests.

Three caveats deserve emphasis. First, the couples panel conditions on marital survival and husband linkage, creating positive selection. The within-couple gains may overstate the experience of all married women if dissolved marriages were associated with labor market disruptions. Second, the ABE algorithm links primarily men; wives are tracked indirectly through husbands' households. Women who remarried, became household heads, or left their husbands' households are excluded. Third, the mobilization measure captures Army enlistment only, not the full scope of military service. To the extent that Army enlistment correlates with total mobilization, this measurement error would attenuate the estimates toward zero.

The broader implication concerns how we interpret the effects of large temporary shocks on labor markets. The aggregate data are not wrong; they measure a different object than within-person data. Compositional change—driven by cohort replacement, mortality, migration, and marital turnover—can generate aggregate trends that diverge substantially from individual trajectories. The within-person trend shows larger gains than the aggregate, not smaller. The development of large-scale longitudinal linkage projects, exemplified by the Census Linking Project, opens the possibility of revisiting canonical questions in economic history with individual-level data that previous generations of researchers could not access.

The war reshaped the American economy and the American family. For the individual women tracked through their husbands across census waves, the decade brought substantial gains in labor force participation—gains that exceeded what aggregate data reveal. The lasting transformation of women's economic lives was driven by broad structural forces operating across all states and all levels of mobilization, not by the specific intensity of the wartime shock. The within-person evidence makes this conclusion precise, quantifiable, and—for the first time—directly observable in the historical record.

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A1. Data Construction Details

A1.1 Census Linking Project Crosswalk

The Census Linking Project (CLP) crosswalk files (Abramitzky et al., 2025) were obtained from the Harvard Dataverse (doi:10.7910/DVN/DXHTEQ). The CLP provides pre-computed person-level crosswalks linking individuals across U.S. decennial censuses from 1850 to 1950. This paper uses the 1940–1950 crosswalk.

The ABE linking algorithm proceeds as follows:

1. **Blocking:** Records are grouped by sex, birthplace (state or country), and approximate birth year (±2 years). Only records within the same block are compared.

2. **Name matching:** Within each block, records are compared on first name (accounting for nicknames and spelling variants) and surname. The algorithm requires exact or near-exact name matches.

3. **One-to-one matching:** When multiple candidates exist, the algorithm selects the unique best match. Ambiguous cases (multiple equally good candidates) are dropped to minimize false positives.

4. **Conservative filtering:** Additional filters remove links where age discrepancies exceed thresholds or where name similarity scores fall below cutoffs.

Because the algorithm matches on surnames, it links primarily men. Women who marry between census waves change surnames, making them effectively unlinkable. In the 1940–1950 crosswalk restricted to individuals aged 18–55 in 1940, only 88 women are linked (compared to over 14 million men). This motivates the couples panel design.

A1.2 Constructing the Couples Panel

The couples panel links wives across census waves through their husbands' households. The procedure:

1. Identify all married men in the 1940 census who are successfully linked to 1950 via the CLP crosswalk.

2. In the 1940 census, identify the spouse of each linked man using the RELATE variable (spouse = relationship to head).

3. In the 1950 census, locate the linked man's household using his SERIAL number.

4. Within the 1950 household, identify the spouse.

5. Record the wife's labor market outcomes (LABFORCE, EMPSTAT, OCC1950, OCCSCORE) in both 1940 and 1950.

This procedure yields 5,564,191 matched couples. The main sources of attrition are: (a) husbands not linked by CLP (∼79%); (b) couples no longer married in 1950 (divorce, death, separation); (c) wife not identifiable as spouse in 1950 household.

An important limitation of this household-based linkage is that the wife identified in the 1950 household may not be the same person as the wife in 1940. If the husband remarried between censuses, the "wife" in 1950 is a different individual, and the within-couple first difference confounds spousal turnover with genuine within-person change. To assess the magnitude of this concern, I construct an age-consistency check: the wife's age in 1950 should equal her 1940 age plus 10. I flag couples as "age-verified" when $|\text{age}_{1950} - \text{age}_{1940} - 10| \leq 2$, allowing for minor enumeration error. Approximately 85.9% of couples pass this check, suggesting that the vast majority of household-linked wives are the same individual across censuses. Results are robust to restricting the sample to age-verified couples (see Section 7).

A1.3 IPUMS Census Variables

The following IPUMS USA variables are used in the analysis, extracted from the full-count (100%) databases for 1940 and 1950 (Ruggles et al., 2024):

- `SERIAL`: Household serial number (used to link wives through husbands' households)
- `STATEFIP`: State FIPS code (identifies state of residence at census date)
- `YEAR`: Census year (1940 or 1950)
- `SEX`: Sex (1 = male, 2 = female)
- `AGE`: Age at time of census enumeration
- `RACE`: Race (1 = White, 2 = Black, 3+ = Other/Multiple)
- `BPL`: Birthplace (state or country of birth)
- `MARST`: Marital status (1–2 = married, 3 = separated, 4 = divorced, 5 = widowed, 6 = never married)
- `LABFORCE`: Labor force status (1 = not in labor force, 2 = in labor force)

- `EMPSTAT`: Employment status (1 = employed, 2 = unemployed, 3 = not in labor force)
- `OCC1950`: Occupation, 1950 classification
- `OCCSCORE`: Occupational income score
- `SEI`: Duncan Socioeconomic Index
- `EDUC`: Education (harmonized categorical variable)
- `RELATE`: Relationship to household head (1 = head, 2 = spouse, 3+ = other)
- `URBAN`: Urban/rural status
- `FARM`: Farm status
- `PERWT`: Person sampling weight (equals 1 for full-count data)

A1.4 CenSoc Enlistment Records

The CenSoc WWII Army Enlistment Records dataset (Version 3.0) was obtained from the Harvard Dataverse (doi:10.7910/DVN/ZFVVNA). The original dataset contains approximately 8.3 million Army enlistee records. After restricting to records with valid state FIPS codes, approximately 2.6 million records remain.

Each record contains the enlistee's state of residence at enlistment, enlistment date, branch of service, and limited demographic information. I aggregate to the state level and compute:

$$\text{Mob}_s = \frac{N_s^{\text{enlist}}}{N_s^{\text{male 18-44}}} \tag{8}$$

where $N_s^{\text{male 18-44}}$ is the number of males aged 18–44 in state s from the 1940 full-count census. The mobilization rate is then standardized to mean zero and unit standard deviation.

A1.5 Variable Construction

From the raw IPUMS variables, I construct:

- **Labor force participation**: $\text{in_lf} = \mathbb{I}[\text{LABFORCE} = 2]$
- **Employed**: $\text{employed} = \mathbb{I}[\text{EMPSTAT} = 1]$
- **Married**: $\text{married} = \mathbb{I}[\text{MARST} \leq 2]$
- **Household head**: $\text{head} = \mathbb{I}[\text{RELATE} = 1]$

- **Urban**: is_urban = $\mathbb{I}[\text{URBAN} = 2]$
- **Farm**: is_farm = $\mathbb{I}[\text{FARM} = 2]$
- **Female**: female = $\mathbb{I}[\text{SEX} = 2]$
- **Education years**: Approximate years from IPUMS EDUC categories
- **Race categories**: White (RACE = 1), Black (RACE = 2), Other
- **Mobilization quintile**: Based on quintiles of the state-level mobilization rate distribution
- **Mover**: mover = $\mathbb{I}[\text{statefip}_{1940} \neq \text{statefip}_{1950}]$

All within-person changes are computed as $\Delta X_i = X_{i,1950} - X_{i,1940}$.

A2. Comparison with Acemoglu, Autor, and Lyle (2004)

Table A10: Comparison of Estimation Approaches

Feature	This Paper	ACL (2004)
Census data	Full-count (100%)	Full-count (selected years)
Linkage	Census Linking Project (Abramitzky et al., 2025)	None (repeated cross-section)
Design	Within-person first-difference	Repeated cross-section
Unit of analysis	Individual (men), Couple (wives)	State
Individual heterogeneity	Absorbed by first-differencing	Not addressed
Mobilization measure	CenSoc Army enlistees	Selective Service inductions
Denominator	Full-count male pop 18–44	Full-count male pop 18–44
Panel sizes	14.0M men, 5.6M couples	48 states
Decomposition	Within-person vs compositional	Not available
Inference	State-clustered SE + RI	State-level robust SE

A3. Additional Robustness Results

A3.1 Linkage Rate by State

Linkage rates vary across states due to differences in name distributions, residential stability, and census enumeration quality. To assess whether differential linkage rates bias the results, I compute the linkage rate for each state and test whether it correlates with mobilization intensity. If high-mobilization states have systematically different linkage rates, the within-person estimates could reflect linkage selection rather than genuine treatment effects.

A3.2 Age-Specific Effects

To further probe the mechanism, I estimate the within-couple specification separately for three age groups (18–30, 31–45, 46+ in 1940). Younger wives were more likely to be wartime labor market entrants and thus more susceptible to both the positive wartime shock and postwar displacement. Older wives had more established household responsibilities and may have been less responsive to wartime labor demand. The results (Section 6, Heterogeneity) show that prime-age wives (31–45) had the largest positive mobilization coefficient (+0.0042), while younger and older wives showed null effects.

A3.3 Sensitivity to Linkage Quality

Following Bailey et al. (2020), I assess sensitivity to potential linkage errors by comparing results across subgroups with different linkage rates. If the results are driven by false positive matches (incorrectly linked individuals who appear to have large within-person changes because they are actually different people), restricting to higher-quality matches should attenuate the coefficient. The consistency of the mobilization coefficient across demographic subgroups with very different linkage rates provides reassurance that linkage errors are not driving the findings.

A4. Replication Instructions

All analysis is conducted in R. The replication archive contains seven numbered R scripts:

- 00 `packages.R` — Load packages, set themes, configure parallel processing, load API keys
- 01 `fetch_data.R` — Download IPUMS full-count extracts (1940, 1950) via the IPUMS API; download CenSoc enlistment data from Harvard Dataverse; download Census Linking Project crosswalk from Harvard Dataverse; download county shapefiles
- 02 `clean_data.R` — **Census Linking Project longitudinal linking**: merge CLP crosswalk with IPUMS full-count data, validate links (age/sex/birthplace consistency), construct individual panel (men) and couples panel (wives via husband's SERIAL), compute within-person changes, construct state mobilization rates, linkage diagnostics
- 03 `main_analysis.R` — Within-person first-difference regressions (men: M1–M4, wives: W1–W4), occupation results (M5, W5–W6), husband-wife dynamics, formal decomposition, state-level cross-validation, heterogeneity analysis by race, pre-war LF status, and age

04 `robustness.R` — Oster coefficient stability bounds, randomization inference (1,000 permutations), HC2/HC3 standard errors, leave-one-out state influence, placebo test on older women, ANCOVA specification, quintile treatment, trimmed sample

05 `figures.R` — Generate all figures (mobilization map, within-person by quintile, decomposition, binned scatter, RI distribution, heterogeneity forest, LOO influence, LF transitions, husband-wife dynamics)

06 `tables.R` — Generate all tables (summary statistics, men's results, wives' results, occupation, husband-wife dynamics, decomposition, state-level, robustness summary, linkage balance)

Scripts must be run in order from `00` through `06`. Data acquisition (`01`) requires an IPUMS API key and CLP access from Harvard Dataverse. All intermediate datasets are saved as `.rds` files in the `data/` subdirectory.

Hardware requirements: The full-count data for two censuses contains approximately 280 million person-records. With `data.table` and selective column loading, peak RAM usage is approximately 30–40 GB. A machine with at least 64 GB RAM is recommended. The randomization inference step (1,000 permutations) benefits from multiple cores.

Software requirements: R 4.2+, with packages: `data.table`, `fixest`, `ggplot2`, `sf`, `ipumsr`, `kableExtra`, `sandwich`, `arrow`. All packages are installed automatically by `00_packages.R`.

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Project Repository: https://github.com/SocialCatalystLab/ape-papers